

Toshiba Bi-CD Integrated Circuit Silicon Monolithic

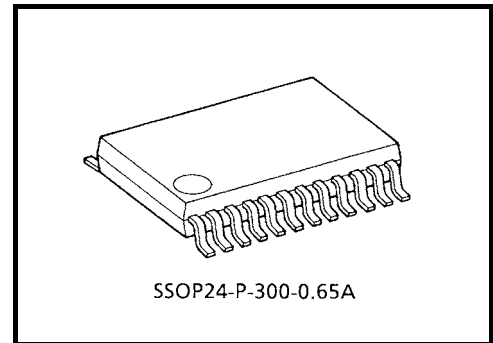
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Driver IC for Dual DC motor

TB6612FNG is a driver IC for DC motor with output transistor in LD MOS structure with low ON-resistor. Two input signals, IN1 and IN2, can choose one of four modes such as CW, CCW, short brake, and stop mode.

Features

- Power supply voltage ; $V_M=15V$ (Max.)
- Output current ; $I_{out}=1.2A(ave) / 3.2A$ (peak)
- Output low ON resistor ; 0.5Ω (upper+lower Typ. @ $V_M \geq 5V$)
- Standby (Power save) system
- CW/CCW/short brake/stop function modes
- Built-in thermal shutdown circuit and low voltage detecting circuit
- Small faced package (SSOP24 : 0.65mm Lead pitch)
- Response to Pb free packaging



質量: 0.14 g (標準)

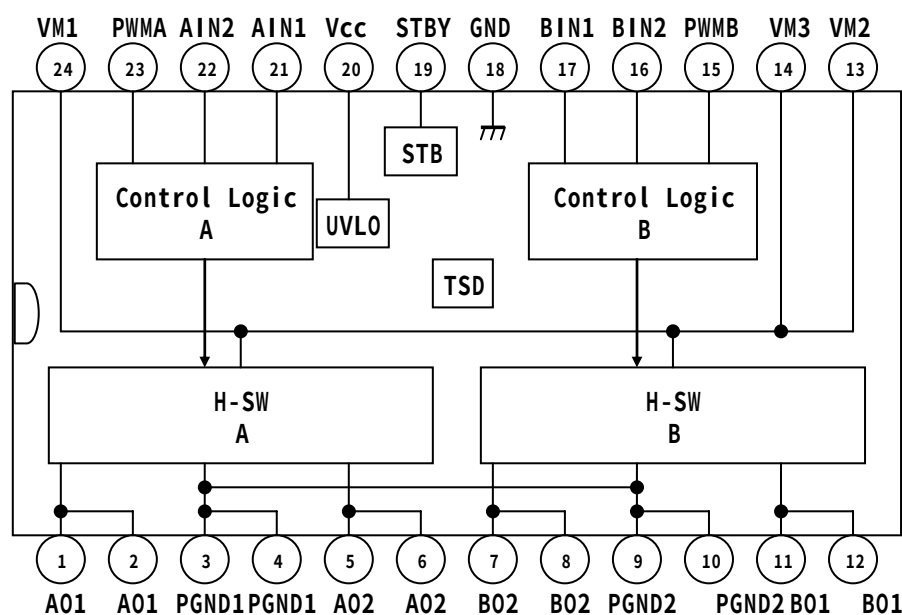
- * This product has a MOS structure and is sensitive to electrostatic discharge. When handling this product, ensure that the environment is protected against electrostatic discharge by using an earth strap, a conductive mat and an ionizer. Ensure also that the ambient temperature and relative humidity are maintained at reasonable levels.

The TB6612FNG is a Pb-free product.
The following conditions apply to solderability:

***Solderability**

1. Use of Sn-37Pb solder bath
*solder bath temperature = 230°C
*dipping time = 5 seconds
*number of times = once
*use of R-type flux
2. Use of Sn-3.0Ag-0.5Cu solder bath
*solder bath temperature = 245°C
*dipping time = 5 seconds

Block Diagram



Pin Functions

Pin NO.	Symbol	I/O	Remarks
1	AO1	O	chA output1
2	AO1		
3	PGND1	—	Power GND 1
4	PGND1		
5	AO2	O	chA output2
6	AO2		
7	BO2	O	chB output2
8	BO2		
9	PGND2	—	Power GND 2
10	PGND2		
11	BO1	O	chB output1
12	BO1		
13	VM2	—	Motor supply (2.5V~13.5V)
14	VM3		
15	PWMB	I	chB PWM input / 200k Ω pull-down at internal
16	BIN2	I	chB input2 / 200k Ω pull-down at internal
17	BIN1	I	chB input1 / 200k Ω pull-down at internal
18	GND	—	Small signal GND
19	STBY	I	“L”=standby / 200k Ω pull-down at internal
20	Vcc	—	Small signal supply (2.7V~5.5V)
21	AIN1	I	chA input1 / 200k Ω pull-down at internal
22	AIN2	I	chA input2 / 200k Ω pull-down at internal
23	PWMA	I	chA PWM input / 200k Ω pull-down at internal
24	VM1	—	Motor supply (2.5V~13.5V)